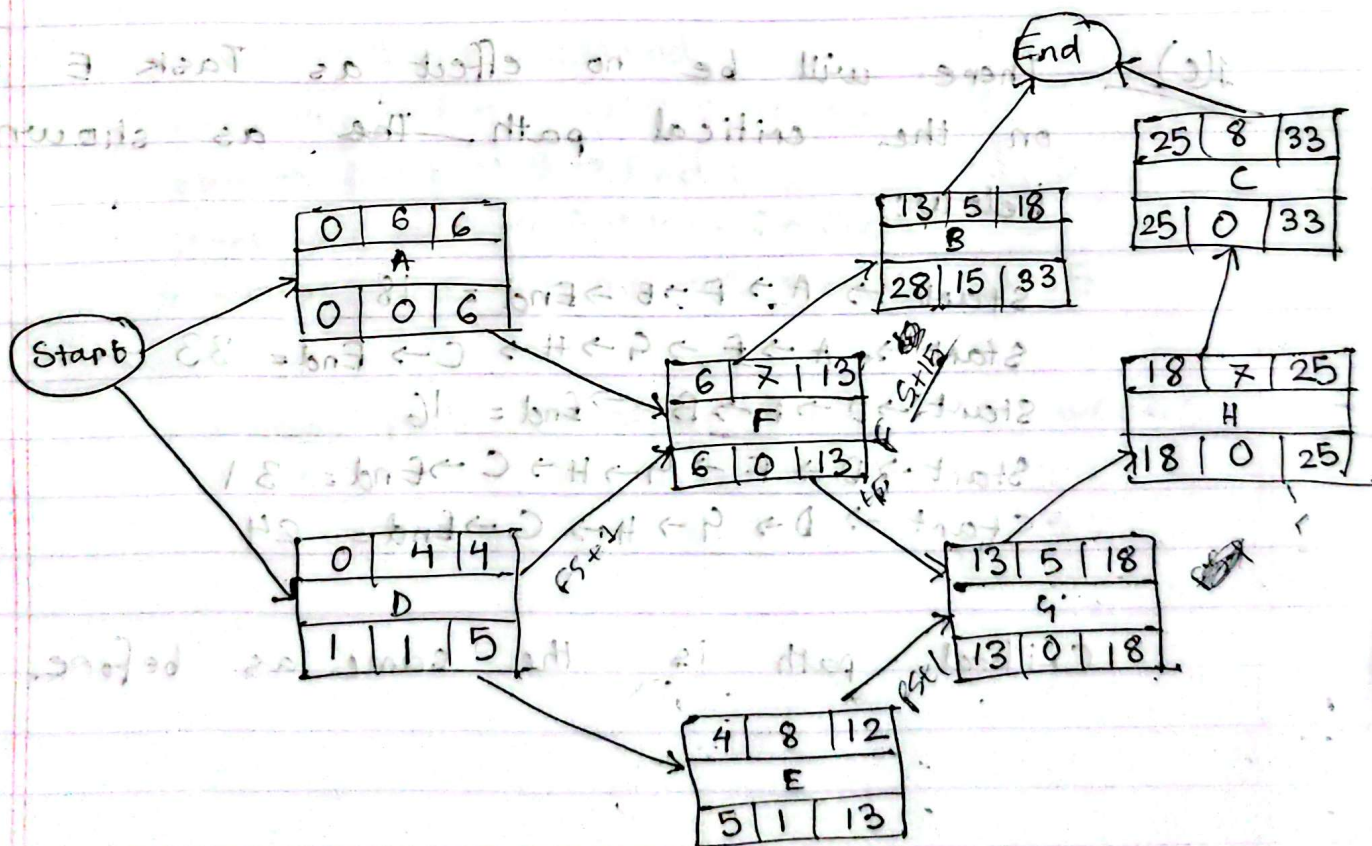


Standard deviation of PERT, $\sigma = \left(\frac{P-O}{6} \right)$

19

1(a)

		3 point $\left(\frac{O+P+M}{3} \right)$	PERT $\left(\frac{O+4P+M}{6} \right)$
✓ Task A	Start	6	5.5
✓ Task B	Task F	5	4.5
✓ Task C	Task H	8	8
✓ Task D	Start	4	4
✓ Task E	Task D	8	7.5
✓ Task F	Task A, D	7	7
✓ Task G	Task E, F	5	4.5
✓ Task H	Task G	7	6.5
End	Task C, B	0	0



Start $\rightarrow$ A $\rightarrow$ F $\rightarrow$ B $\rightarrow$ End = 18

Start $\rightarrow$ A $\rightarrow$ F $\rightarrow$ G $\rightarrow$ H $\rightarrow$ C $\rightarrow$ End = 33 ✓

Start $\rightarrow$ D $\rightarrow$ F $\rightarrow$ B $\rightarrow$ End = 16

Start $\rightarrow$ D $\rightarrow$ F $\rightarrow$ G $\rightarrow$ H $\rightarrow$ C $\rightarrow$ End = 31

Start $\rightarrow$ D $\rightarrow$ E $\rightarrow$ G $\rightarrow$ H $\rightarrow$ C $\rightarrow$ End = 32

4(a)

33 weeks

4(b)

Task B $\rightarrow$ 15

Task D $\rightarrow$ 1

Task E $\rightarrow$ 1

4(c)

There will be no effect as Task E is not on the critical path as shown below:

Start $\rightarrow$ A $\rightarrow$ F $\rightarrow$ B $\rightarrow$ End = 18

Start $\rightarrow$ A $\rightarrow$ F $\rightarrow$ G $\rightarrow$ H $\rightarrow$ C $\rightarrow$ End = 33 ✓

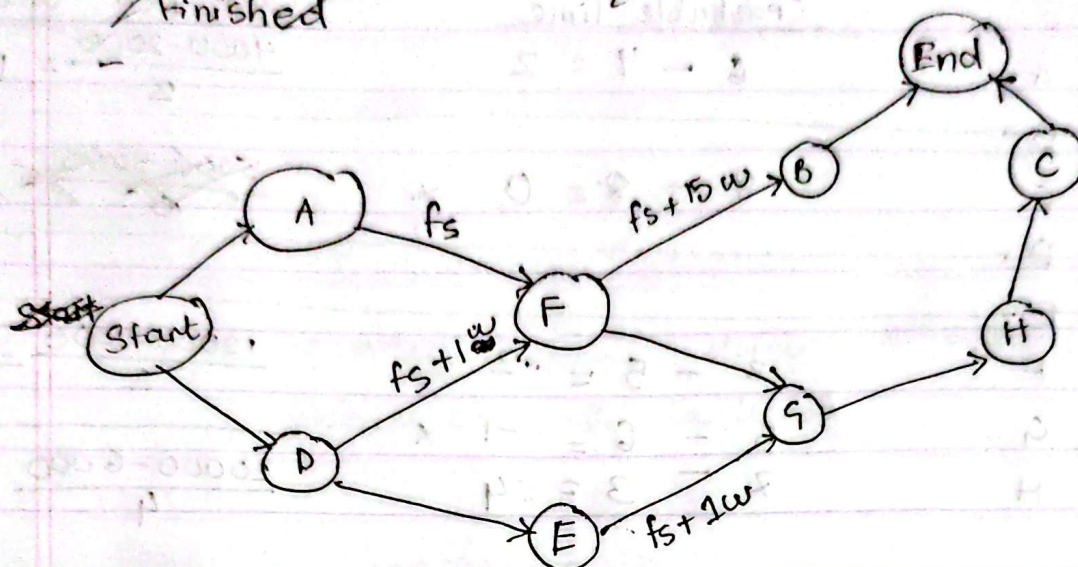
Start $\rightarrow$ D $\rightarrow$ F $\rightarrow$ B $\rightarrow$ End = 16

Start $\rightarrow$ D $\rightarrow$ F $\rightarrow$ G $\rightarrow$ H $\rightarrow$ C $\rightarrow$ End = 31

Start $\rightarrow$ D $\rightarrow$ G $\rightarrow$ H $\rightarrow$ C $\rightarrow$ End = 24

Critical path is the same as before.

1(d) - Task F will start after Task A and D are finished



1(e)

Start $\rightarrow$ A $\rightarrow$ F $\rightarrow$ B $\rightarrow$ End = 17

Start $\rightarrow$ A $\rightarrow$ F $\rightarrow$ G $\rightarrow$ H $\rightarrow$ C $\rightarrow$ End = 31.5

Start $\rightarrow$ D $\rightarrow$ F $\rightarrow$ B $\rightarrow$ End = 15.5

Start $\rightarrow$ D $\rightarrow$ F $\rightarrow$ G $\rightarrow$ H $\rightarrow$ C $\rightarrow$ End = 30

Start $\rightarrow$ D $\rightarrow$ E $\rightarrow$ G $\rightarrow$ H $\rightarrow$ C $\rightarrow$ End = 30.5

So critical path duration is 31.5 weeks.

Crashing

only for critical path

	Crashable Time	Cost per week
Task A	$6 - 4 = 2$	$\frac{4000 - 2000}{2} = 1000$
Task B		
Task C	$8 - 8 = 0 \times$	$\frac{5000 - 5000}{0}$
Task D		
Task E		
Task F	$7 - 5 = 2$	$\frac{12000 - 8000}{2} = 2000$
Task G	$5 - 6 = -1 \times$	
Task H	$7 - 3 = 4$	$\frac{20000 - 6000}{4} = 3500$

- can't crash Task C & G

- possible crashing activities for 4 weeks early:

task A & F $\rightarrow 1000 \times 2 + 2000 \times 2 = 6000$

task H $\rightarrow 3500 \times 4 = 14000$

Since, crashing task A & F $<$ task H so we should crash task A & F (Ans)

lecture 5.2, slide - 42

2(a)

Cost of conformance

↳ money spent during the project to avoid failures

Cost of nonconformance

↳ money spent during & after the project because of failures

Cost of conformance

annual investment = \$100,000

∴ CoC = \$100,000

Cost of non conformance

annual defects = $60 \times 4 = 240$

cost to fix post-release defects = $5,000 \times 240$
~~= \$900,000~~
= \$1,200,000.

customer support expense = $20,000 \times 240$
= \$4,800,000

CoNC = $1,200,000 + 4,800,000$
= \$6,000,000 (Ans)

Since the cost of nonconformance is very high compared to the cost of conformance, the company can improve their QA area in the following ways:

- ① test the product rigorously through automated testing processes
- ② perform regression testing after fixing defects each time to prevent further defects
- ③ use customer feedback to detect and solve issues early
- ④ invest more in prevention activities like better training, improve requirements gathering etc.
- ⑤ use defect metrics to track & improve quality over time

Lecture 5 part 2, slide -11

Cost estimation techniques

More Precise

- Parametric
- Bottom-up
- Three-point

Less Precise

- Analogous

Analogous vs Parametric

	<u>Analogous</u>	<u>Parametric</u>
<u>Method</u>	- use previous historic data of similar projects	- use formula on historic data of previous project
<u>when</u>	- early phases of the project when only high level data can be used	- when detailed data is available
<u>Utility</u>	only when data from a similar project is available	only when a formula, algorithm or statistical model can be devised
<u>Accuracy</u>	Usually less accurate - depends on the expertise & experience	Usually more accurate - depends on the accuracy of available data and chose model.
<u>Effort</u>	Usually less costly & time consuming	Usually more costly & time consuming

Lecture 7, slide 35

Power of a project manager

~~FRPER~~ P R E F R

- ① Penalty (Coercive) - ability to penalize team members
- ② Reward - this power stems from giving reward
- ③ Expert - comes from being the technical expert or project management expert
- ④ Formal (legitimate) - power is based on the position of the project manager.
- ⑤ Referent - power comes from charisma and fame. This power comes from another person liking the project manager, respecting him & wanting to be like him.

P R E F R - Referent.
Penalty Reward Expert Formal

Lecture 6, slide 40

Strategies for threats

AMTAC

① Avoid - if you can prevent it from happening, it definitely won't hurt your project.

② Mitigate - if you can't avoid the risk, then mitigate it. This means taking some sort of action which will cause it to do as little damage to your project as possible.

③ Transfer - pay someone to accept the risk. The most common way is to buy insurance.

④ Accept - if you can't do anything to avoid the risk and you can't reduce its impact then accepting it is your only choice.

(3)(a) transfer - buy insurance

Mitigate - adapt project to market demands using agile methodology

Mitigate - perform early testing to check functionality & performance

~~Am~~ Avoid - use updated technology

Mitigate - use reserves & monitoring to reduce impact

[lec 8 slide 40]

69:29

3(b)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
QA1	4	4	3	3	2	5	5	4	5	5	1	1	10	8	1
QA2	4	4	3	3	2	3	5	5	5	5	1	1	4	8	1
QA3	4	2	3	2	3	3	4	4	6	5	3	1	3	4	2
QA4															
Total bugs for each day	12	10	9	8	7	11	14	13	16	15	5	3	17	20	4

$$\mu = \frac{164}{15} = 10.9$$

$$\sigma = \sqrt{\frac{\sum (\mu - x)^2}{n}} = \sqrt{\frac{350.95}{15}} = 4.8$$

$$\mu + \sigma = 15.7$$

$$\mu - \sigma = 6.1$$

$$\mu + 2\sigma = 20.5$$

$$\mu - 2\sigma = 1.3$$

$$\mu + 3\sigma = 25.3$$

$$\mu - 3\sigma = -3.5$$

1 → 1 above 3σ

2 → 8 or more on one side of CL

3 → 4/5 points in zone B

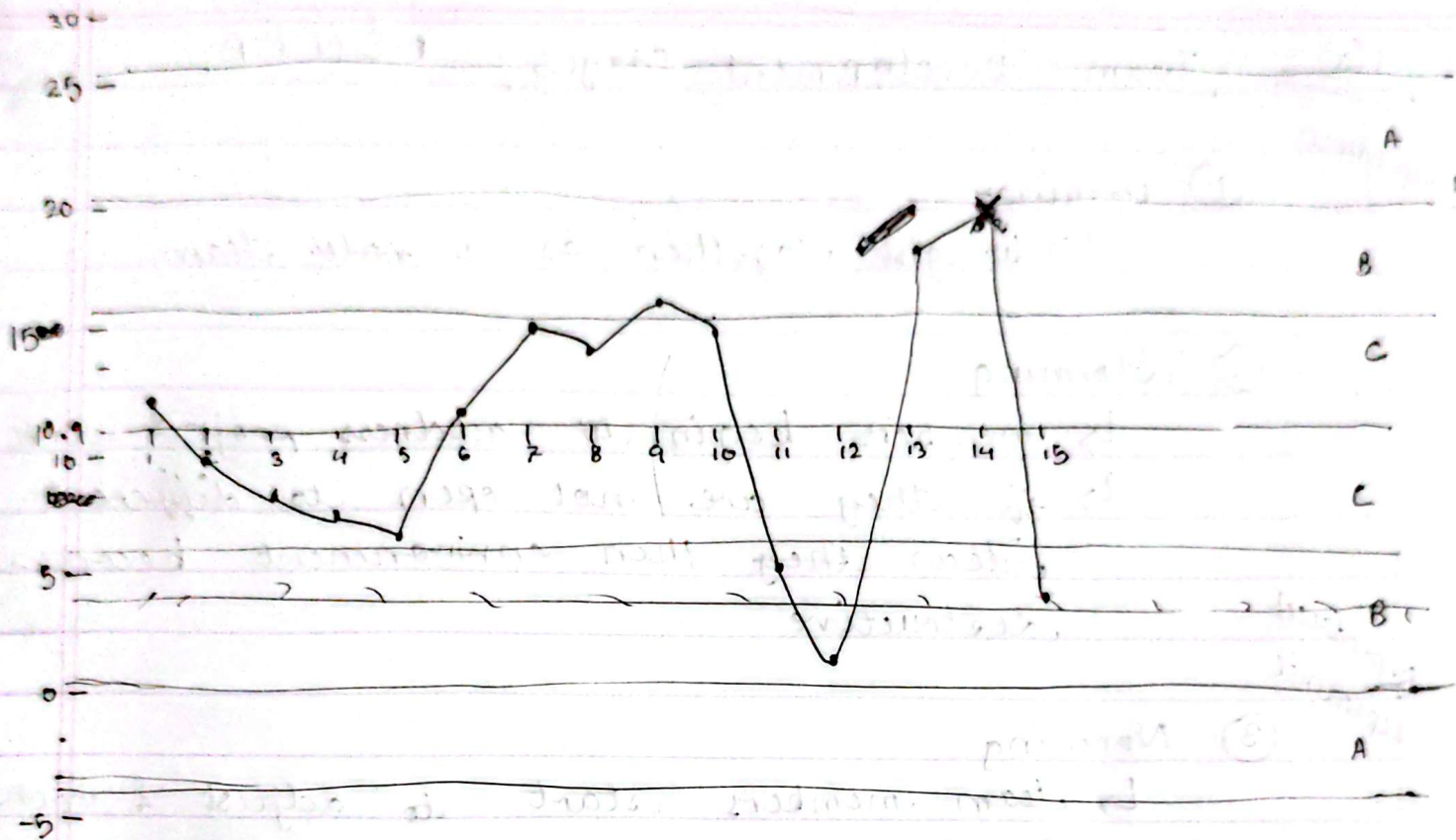
4 → 6 or more points in inc/dec

5 → 2/3 points in zone A

6 → 14 points zig zag / up down

7 → contr repeating pattern

11 3 4 8 4 7



point 10 - 14 indicates Rule-3 out of control signal that is 8 → 4 out of 5 points consecutive points are in zone B or above.

points 11, 12, 13 & 14 → in zone B

not same side all 4 points.

So no rules broken!

Lec 7, slide 31

for some reason
personal assistant

4(a)

Team Development Stages

F S N P A

Independence

① forming

↳ brought together as a ~~value~~ team

② Storming

↳ members begin to address project work

↳ if they are not open to different ideas then environment becomes destructive

Dependence/
Independence

③ Norming

↳ team members start to adjust & work together

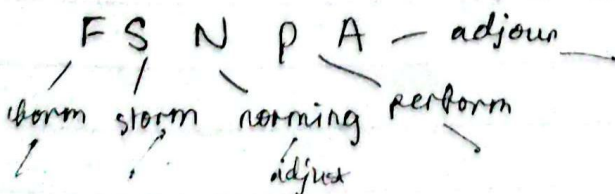
④ Performing

↳ team members function together as a well organized unit

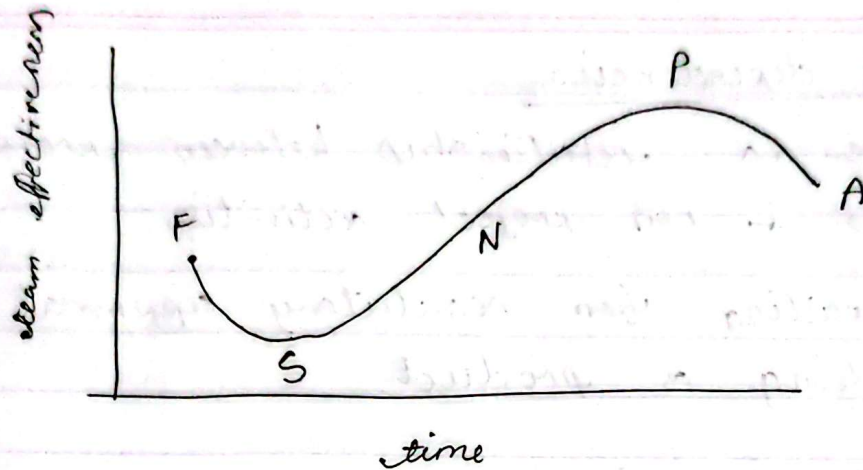
Independence
Dependence

⑤ Adjourning

↳ completes the project work & moves on from the project



4(a)



4(a) Performing stage [lect 7, slide 31]

Effectiveness $\rightarrow$ graph

4(b) Dependency Determination [lec 5.1, slide 76]
M D E I

① Mandatory Dependencies

- contractually required in the nature of the work called hard logic
- e.g. must design database schema before coding queries

② Discretionary Dependencies

- defined by project management team
- known as "soft logic", "~~dis~~ preferred logic" & "preferential logic".
- e.g. you prefer to finish the UI before backend even if both can go in parallel.

③ External dependencies

- involve a relationship between project activities & non project activities
- e.g. waiting for regulatory approval before launching a product.

④ Internal dependencies

- involve a ~~to~~ precedence ^{relationship} between project activities and are generally within the project team's control
- e.g. QA team can begin testing only after development team is done.

4(c)
Sec 8,
slide 16

- track progress in real-time → show actual remaining progress work vs. ideal remaining work
- identify early warning signs → if actual line is ^{above} ~~behind~~ ideal line then team is behind schedule
- forecast completion → whether team will finish on time based on current activity velocity
- promote transparency & accountability → chart visible to team & stakeholders

- manage scope changes → show sudden increase in burn-down chart if work added

- motivate team

5(a)

$$BAC = 100,000$$

$$EV = 40,000$$

$$AC = 60,000$$

$$PV = 50,000$$

$$MV \text{ of remaining work}$$

$$= BAC - EV$$

$$= 60,000$$

$$SPI = EV/PV = 0.80$$

$$CPI = EV/AC = 0.67$$

$$ETC = \frac{MV \text{ of remaining work}}{CPI}$$

$$= \frac{60,000}{0.67} \times 0.67 = 90,000$$

$$EAC = ETC + AC$$

$$= 90,000 + 60,000 = 150,000$$

$$TCPI_B = \frac{BAC - EV}{BAC - AC} = \frac{60,000}{100,000 - 60,000}$$

$$= 1.5 \text{ (Ans.)}$$

Lec 7, slide -40

PATH CSP

5(b)Sources of conflict:

- ① project priorities
- ② administrative procedures
- ③ technical options and performance trade-offs
- ④ human resources
- ⑤ cost
- ⑥ schedules
- ⑦ personality conflict

Next slideHow to resolve conflicts

W/A

S/A

C/R

F/D

WSCFC

- ① withdraw / Avoid
↳ one party chooses not to deal with
- ② Smooth / Accommodate
↳ you try to keep the peace by agreeing with the other person even if you don't fully agree
- ③ Compromise / Reconcile
↳ each party gives up part of their demands to reach a middle ground.
- ④ Force / Direct
↳ one party imposes their will over the other

⑤ Collaborate / Problem Solve

↳ both parties work together to find a win-win solution

• Most conflicting process

- Initiating → project charter
- Planning → project management plan
- Execution → develop team
 - manage team
 - manage stakeholder engagement
 - manage communication

5(c) [lecture 6, slide 26]

Risk parameters

① Urgency

↳ short period = high urgency

② Proximity

↳ short period = high proximity

③ Dormancy

↳ short period = low dormancy

④ Manageability

↳ high = easy to manage

⑤ Controllability

↳ high = easy to control

⑥ Detectability

↳ high = easy to detect

⑦ Connectivity

↳ high = highly connected

⑧ Strategic Impact

↳ high = major effect

⑨ Proximity

↳ high = highly significant

① "The development timeline is tight, with a need for rapid response to any security ~~vulner~~ vulnerabilities to ensure effective risk mitigation"

— Urgency

② "Integration issues with third-party banking platforms could directly impact project objectives, requiring impact ~~project~~ ~~objective~~ immediate attention & resolution."

— Proximity (high)

③ "Potential security breaches may go undetected initially, leading to long-term financial and reputational consequences."
— Dormancy

④ "Controlling outcomes, especially in response to external factors like evolving regulatory requirements, presents ongoing challenges
— Controllability

⑤ "Potential security breaches may go undetected..."
— Detectability (low)

⑥ "Interconnected nature of project risks means that a failure in one area could cascade into multiple other risks, affecting project success."
— Connectivity (high)

⑦ "Stakeholders, including customers and regulators, perceive security and reliability as paramount."
— Propinquity (high)

Lecture 9, slide 4 & 5

G(a)

Types of project closing

ASIE

① Addition

↳ project evolved into ongoing operation

② Starvation

↳ when resources are cut off from the project

③ Integration

↳ People, equipment and other resources are distributed to other areas.

④ Extinction

↳ project has been completed & accepted by stakeholders

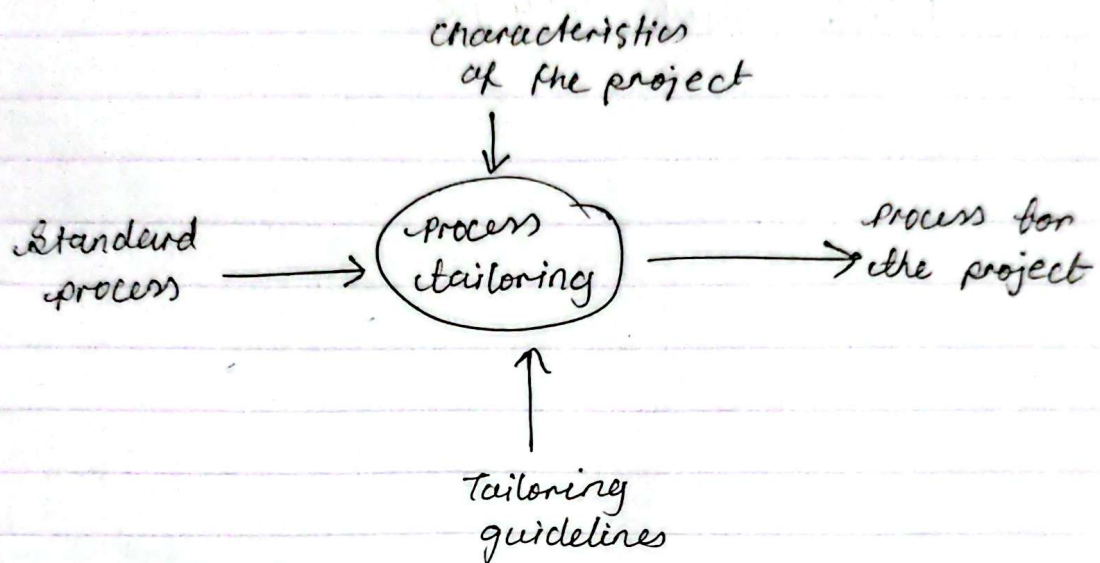
Why - do we need to close a project?

- successful completion of the project
- changes in organisational priorities
- lack of funding
- technological changes
- failure to meet project goals
- legal or regulatory issues
- mergers or acquisitions
- external factors

Lec-9, slide 15

6(b) process tailoring

Adjusting a previously defined process of an organization to obtain a process that is suitable for the business or technical needs of a project.



Lec-6, slide 35

6(c) Vendor X

on time	0.5	0
being late	0.3	-40,000
early	0.2	+20,000
quality issue	0.2	-20,000

$$\begin{aligned} EMV_X &= 0.5 \times 0 + 0.3(-40,000) + \\ & 0.2 \times (20,000) + 0.2(-20,000) \\ &= -12,000 \end{aligned}$$

Vendor Y

on time	0.3	0
being late	0.4	-40,000
early	0.3	+20,000
quality issue	0.15	-25,000

$$\begin{aligned} EMV_Y &= 0.3 \times 0 + 0.4(-40,000) + \\ & 0.3(20,000) + 0.15(-25,000) \\ &= -13,250 \end{aligned}$$

So Vendor X

18

1(a)

Lect 5, slide 21

6(d)

[Maybe Lec 2, slide 31]

PMI Talent Triangle

